

Automated Aquaponic Garden PLC Using a Web-based Interface

EE-4910
GROUP 11

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Device Purpose

- Increase accessibility of small-scale aquaponic systems
- Provide a tool for home gardeners to automate their aquaponic yields
- Create a schedulable system of powered appliances
- Allow remote access and monitoring of aquaponic operation
- Improve efficiency of agricultural water consumption
- Support self-sufficiency of private citizens

Design Scope

- Access 8 independently controlled electrical outlets through a micro-computer that manages the outlets and hosts a web page, allowing verified users to direct power to the aquaponic electrical components
- Allow users to run schedules on equipment, store and query data and logs, and optionally import open-source agricultural databases
- Develop and build a dynamic, indoor, automated aquaponic garden that uses a web-based interface to automate the system components

Market Analysis

- A quick amazon search for “aquaponic systems” yields many different products, such as this \$1500 system:
- Our budget is well below this price, able to grow up to 107 plants (rather than 72) and gives a slightly larger growing area.
- Many additional, smaller scale products are also available that offer timers or emergency shutoffs for power strips



Design Objectives

- Objective Prime: The designed invention becomes as integrated into the modern-day kitchen as an oven, refrigerator, and sink. The designed invention is a leader in automating the aquaponic industry.
- 1: Device allows users to independently operate 8 different outlets using remote access
- 2: Device senses when the livestock tank water level is low and stops operations until water is added to the required level
- 3: Device hosts a webpage that allows users to run automated scheduling for lights, aeration, and water pumps; store growth data; and log system modifications

Design Objectives Cont.

- 4: Device communicates with a server to notify users of new software versions and grow data
- 5: Device includes a prototype aquaponic system to show functionality, using a pipe system that would allow for up to 107 plants:
 - Size: 100x50x110cm/38"x19"x43" (of each pipe)
 - Hole Size: 32mm
 - Tube Diameter: 63mm
 - Width: 19 in
 - Length: 43 in



Design Assumptions

The product will:

- Be powered by a GFCI compliant outlet
- Provide power to 120V, three-prong appliances
- Be used indoors, in a dry environment below 50% humidity
- Be used between 40°F and 100°F
- Be connected to an IEEE 802.11 wireless network
- Be secured to a stable, flat surface with screws
- Be used in conjunction with a fillable and drainable tank
- Be used in compliance with provided instructions

Design Requirements

Device:

- Modify flow independently on 8 electrical sockets
- Host a webpage to allow user interaction
- Store growth and yield data, schedules, and system logs
- Maintain system time to ensure accurate scheduled events

Prototype:

- Use multiple water pumps, aeration devices, and light
- Operate as a growth medium for various vegetation
- Operate as an environment for aquatic livestock
- Manage livestock tank water for distributing to grow beds

Design Requirements Cont.

Webpage:

- Allow assignment of pre-selected or custom appliance types to individual outlets
- Maintain independent daily schedules for individual sockets
- Allow scheduling of outlets based on assigned type
- Allow user settings for maximum and minimum operating water levels
- Notify users via email of unscheduled changes to the system state
- Allow users remote access to scheduling through a wide area network
- Maintain password-verified identification of users with access to the system

Design Constraints

- Device will be built using a Raspberry Pi board with SD card
- A relay board will bridge between the outlets and the GPIO and will be scripted in Perl
- System will have a maximum power consumption of 20 amps
- Prototype will be built using a ~50-gallon storage tank
- Device and prototype system will sum to less than \$200USD in total cost

Design Standards

- ISO/IEC 12207:2008 – Systems and software engineering – Software life cycle processes, ISO, 2008
- ISO/IEC/IEEE 26511:2018 – Systems and software engineering – Requirements for managers of information for users of systems, software, and service, ISO/IEC JTC 1/SC 7, 2018
- NEMA 5/IEC 60906-2:2011 – IEC system of plugs and socket-outlets for household and similar purposes, PEL/23, 2011
- IEEE 802.11ac-2013 – IEEE Standard for Information technology – Telecommunications and information exchange between systems, C/LM, 2013

Design Functionality

- User Interface
 - Allows user interactions with the system
 - Interactions include logging data, setting schedules, monitoring plant growth, configuring associative devices
 - Submits user data to the System API to process
- System API
 - Supplies an intermediary between the user interface, data store, and flow controller
 - Processes data from the User Interface
 - Stores data in the Data Store
 - Actuates the Flow Controller

Design Functionality Cont.

- Flow Controller
 - Receives signal from system API and allows or denies current flow
 - Alters the power being sent to different devices within the Aquaponic System
 - Sends signals via GPIO to individual channels on an 8-channel relay board, altering current flow to the strip board socket
 - Senses water level and shuts off specific sockets when pre-set conditions are not met
- Data Store
 - Stores data necessary to keep system functioning properly, such as system logs and scheduling data
- Aquaponic System
 - Supplies an environment for the growth of aquatic livestock, vegetation, and/or flowers for agricultural purposes

Design Impact

- Cultural :
 - Automated aquaponic systems being used in underdeveloped countries could affect the economy enough to see the country adopt new cultures or reevaluate current cultural norms
 - These changes may be positive or negative
- Economic:
 - Farms could reduce the cost to operate which could lead to more profit
 - Negatively, this larger profit margin could affect the amount of subsidies granted
- Environmental:
 - Lower water usage for agriculture, lower pesticide use, lower fertilizer runoff into public water tables, increase agricultural yield through layering, allows agriculture in barren areas
 - Negative impacts would include trash from bad units; mitigations would include refurbishing units that went bad, stripping materials from bad units, and using safe e-waste methods

Design Impact Cont.

- Global:
 - Enough nutritious food, including some staple foods, for all the family year-round
 - This includes extra food stocks for processing and sale to obtain income, and a reserve for special occasions or emergencies
 - Sales from home garden produce can make a substantial contribution to a family's income
- Public Health:
 - With aquaponic systems, there is less usage of pesticides in general population's diets

Design Impact Cont.

- Public Safety:
 - No use of any toxic chemicals or fertilizers or pesticides
 - A negative impact on growth of the plants is the effect of insect pests; pesticides are not a practical answer in dealing with problems with insects in aquaponic environments
- Public Welfare:
 - With a moderate initial cost and a low upkeep cost, a household aquaponic system can reduce the cost, and increase the supply, of fresh produce to the individual
 - Negative impacts are potentially higher water and electricity utility costs per household
 - Potential mitigations are efficient scheduling suggestions can help reduce the power draw of the system
- Social:
 - Inexpensive aquaponic systems can provide a more accessible and consistent source of healthy produce for the general public
 - Negative impacts are that widely available aquaponic systems could reduce the demand for local and/or organic produce, adversely affecting orchards and farms
 - Potential mitigations are farms could focus more heavily on produce that cannot be generated through aquaponics

User Documentation

This product will be delivered with a paper instruction manual included. The instruction manual will be a packet, written in English, covering the following:

- Explanation of aquaponic process, goals, and benefits
- Safety warnings and safety standards
- Parts list of all provided components
- Parts list of suggested components for building the non-electrical system
- Instructions on both building the provided system and its suggested use
- Instructions on connecting the processor to a wireless network
- Instructions on access the built-in webpage and creating a user account
- Instructions on function and use of the webpage, including troubleshooting, type assignment, scheduling, and settings
- Examples of settings and their applications within the system
- Limited product guarantee
- Contact information for user support

Technical Approach

- Hardware Decisions:
 - Microcomputer: Raspberry Pi Model 3 A+ was chosen due to its wireless capabilities and processing speed
 - Water Sensors: Adafruit and Seed sensors were chosen for their accuracy and ability to easily connect to the circuit board of the computer
 - Relay Board: Elegoo relay board is the most cost-efficient and applicable component of the options for connecting the computer to the electrical sockets
 - Electrical Sockets: Electrical sockets from Legrand fit the criteria best, based on the ability to modify the connection between individual sockets, and a significantly lower price
- Software Decisions:
 - Database: SQLite was chosen because it does not require a server to function, it handles storing and retrieving blob data, and the license is in the public domain
 - Backend: For writing the operational code, which must be able to access the computer, the SQL database, and the webpage, PHP was chosen
 - Frontend: For user interface portion of the software, JavaScript was chosen to create a webpage in the minimal Nevada style

Hardware

- Circuit Board/Single-Board Computer
 - To allow for new feature that enable/use 5 GHZ wireless and increased speed, this design is using Raspberry Pi Model 3 A+, with Raspberry Pi Zero WH (Zero W with Headers) as the runner up due to price
- Water Sensor: Temperature
 - The DS18B20 Digital Temperature Sensor from Adafruit is the most fitting temperature sensor overall. While it is not the least expensive, it does meet the requirements for integration with our Circuit Board and temperature range as specified in the Design Trades
- Water Sensor: Water Level
 - The Grove Water Level Sensor provides a cheap, accurate method of detecting the physical water level at any time. Unlike the float switch, which only senses 3 points (acceptable, high, low), it transmits constant data of the level relative to the sensor
- Relay Board
 - ELEGOO 8 Channel DC 5V Relay Module with Optocoupler, due to best fit with system design
- Power Receptacles
 - Legrand® TradeMaster® 15-Amp Light Almond Duplex Outlet 3232LACC30 due to price; we intend on getting 1 of the 15 Amp USB outlets to direct current to the raspberry pi

Itemized List

- Raspberry Pi Model A 3+ will cost \$25
- DS18B20 Digital Temperature Sensor – Adafruit will cost \$9.95
- Grove Water Level Sensor – Seed will cost \$6.95
- ELEGOO 8 Channel DC 5V Relay Module with Optocoupler will cost \$10.99
- Legrand® TradeMaster® 15-Amp Light Almond Duplex Outlet 3232LACC30 .43 cents individually x 4 will cost \$1.72
- 3-inch PVC Pipe (36 feet) will cost \$14.99 x 4 will cost \$60 (prototype)
- Hub PVC Sewer and Drain Tee \$3.78 individually x 6 will cost \$22.68 (prototype)
- 3-inch Hub Long Turn 90 Degree PVC Sewer and Drain Street Elbow \$3.52 x 12 will cost \$42.24 (prototype)
- PVC Vent Guard \$3.55 individually x 3 will cost \$10.65 (prototype)
- Total: \$190.18 (\$54.61 Automated Aquaponic PLC) (\$135.57 prototype)

Software

- Database
 - SQLite is the optimal choice for this project. Although MariaDB meets many of the same requirements as SQLite, SQLite does not require a server to be running to access a database (database is instead stored in a file), allowing for quicker access and less processing power required. Additionally, SQLite is open-source and highly compatible with the Raspbian operating system
- Backend Programming Language
 - PHP is the language of choice due to it being the only language that satisfied requirement 11, matching the structure of existing solutions to backend software
- Frontend Programming Language
 - JavaScript does not meet requirement 32, however, due to limitations of PHP (i.e., client-side processing), JavaScript will be used
- User Interface
 - Nevada: Minimal One-page HTML Template was the best choice since the time needed to implement the user interface was simple and easy along with it being able to simplify the user interface with the system functions and supports the design of the entire system as well as easy navigation

Tests and Evaluation

- Appropriate output of sockets will be tested under various conditions and states of the webpage
- Appropriate output of sockets will be tested under various water levels
- Web accessibility and persistence will be tested across multiple devices and browsers
- Database accuracy will be tested through direct and automated data insertion
- Proper appliance operation will be tested via visual confirmation

All tests will be performed at the site of construction at Wright State University, using tools available at the facility

Critical Milestones

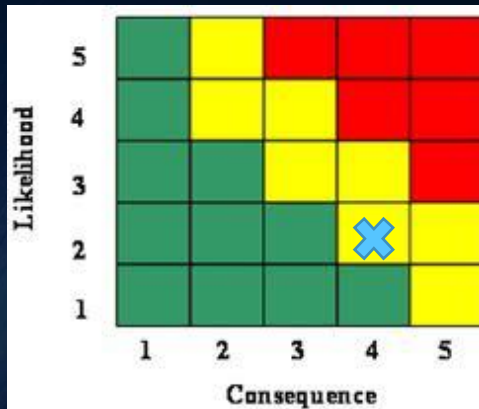
Critical Milestones	Start by:	Done by:
Ordering and receiving parts	April 30, 2021	August 23, 2021
Physical construction of PLC including power strip, outlets, and microprocessor	August 23, 2021	September 6, 2021
Construction of prototype system	August 30, 2021	September 6, 2021
Software development for functional webpage and demonstration of working user interface	August 30, 2021	October 11, 2021
Software development of backend and database, with demonstration of information exchange between user and system	August 30, 2021	October 18, 2021
Complete testing and evaluations of all systems	October 18, 2021	November 14, 2021

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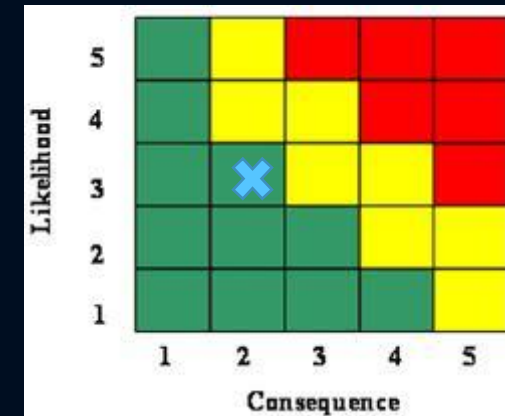
Risks and Mitigations

- Risk 1: Budget and schedule are insufficient to complete construction of prototype system
- Mitigation: Personal contributions and work outside of scheduled hours

Unmitigated



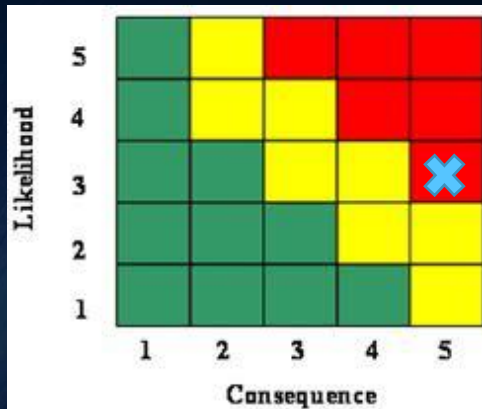
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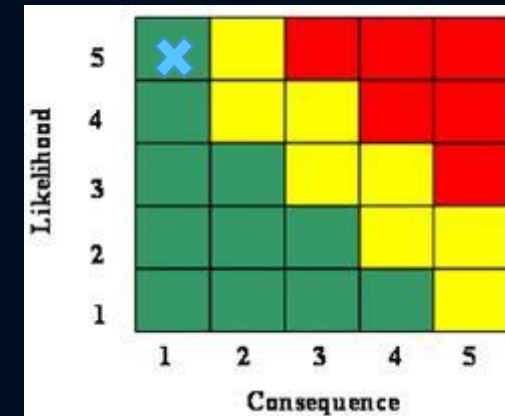
Risks and Mitigations Cont.

- Risk 2: Standards are unable to be met or kept
- Mitigation: Documentation will be provided or sections of design will be reworked

Unmitigated



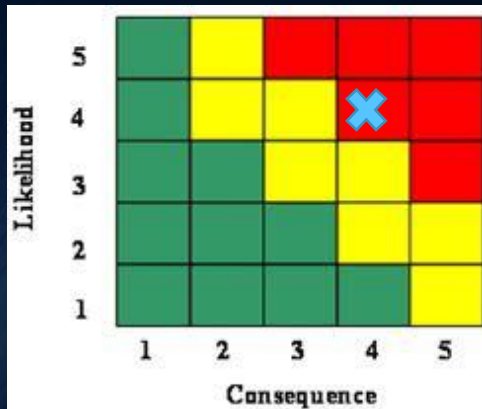
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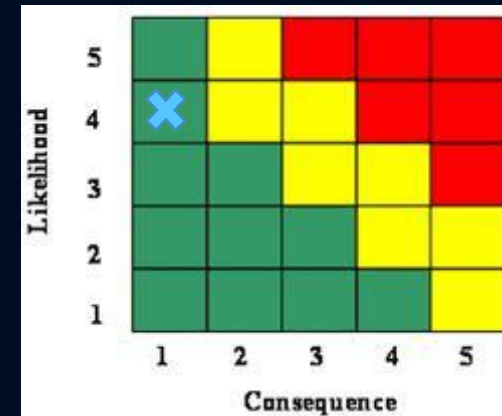
Risks and Mitigations Cont.

- Risk 3: Insufficient knowledge is required for aspects of software development
- Mitigation: Additional research will be done and new sources will be found

Unmitigated



Mitigated



References

- <https://www.instructables.com/Simple-and-intuitive-web-interface-for-your-Raspbe/>
- https://www.bbc.com/future/bespoke/follow-the-food/why-our-food-needs-to-use-less-water/?utm_source=pocket-newtab<https://www.health.harvard.edu/blog/backyard-gardening-grow-your-own-food-improve-your-health-201206294984>
- <http://www.fao.org/in-action/globefish/market-reports/resource-detail/en/c/1263851/>
- https://images-na.ssl-images-amazon.com/images/I/81StZVxm8JL._AC_SL1500_.jpg

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